

Picoclimate XAI: Checkpoint and 1-Month Plan

M2 Internship Meeting

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Goal and Alignment Gap

- Goal: stable, sensor-unit explanations for spatio-temporal clustering.
- Alignment gap: representation, clustering, explanation optimized separately.
- Requirement: explanations in physical units for planners.

Checkpoint: What Is Working

- Mixed-length shapelet dictionaries improve clustering quality in several datasets.
- Stability protocol: multi-seed runs with top-k overlap and surrogate fidelity.
- Bounded sampling makes large traces feasible for repeated evaluation.
- EDA pipeline and report artifacts are reproducible and exportable.

Standard techniques

- Finite differences (Δ , Δ^2 , Δ^3).
- Multi-scale shapelets exist in literature.
- Jaccard overlap for attribution stability is known.

Defensible contributions

- Stability-optimized mixed-length dictionary selection for clustering + XAI.
- Stable explanation protocol for unsupervised clustering in sensor units.
- Four-term joint objective with spatial + temporal penalties.
- Alignment gap taxonomy for planner-facing interpretability.

Core Contribution Framing

- Application-driven method: align representation, clustering, explanation.
- Sensor-unit output contract: consistent narrative across time windows.
- Stability as a first-class metric, not an afterthought.

1-Month Plan (May 21 – June 20)

- Week 1: impute raw slots, rerun $\Delta/\Delta^2/\Delta^3$ baseline.
- Week 2: map indices to physical sensor units in explanations.
- Week 3: expand stability to 10 seeds; finalize tables.
- Week 4: manuscript polish; optional joint-objective prototype.

Immediate Decision

- Priority for this afternoon:
- A) derivative baseline with imputation
- B) sensor-unit mapping for explanations

Risks and Mitigations

- Risk: derivatives degrade clustering after imputation.
- Mitigation: report negative result and favor statistical summaries.
- Risk: runtime for 10-seed sweeps.
- Mitigation: keep bounded sampling and log costs.